

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT2-Constraint-Based Problem Solving

Register Number	192511061
Name	Vidhushini.T.S

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 20%

QUESTIONS:3

Total Marks:30

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- D1 cannot work Night shift
- D2 must work before D3 (Morning < Afternoon < Night)
- Only one doctor per shift
- D3 cannot work Morning
- All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- Movement allowed: Up, Down, Left, Right
- Cost of each move = 1
- Cannot pass through obstacles
- Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed

- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

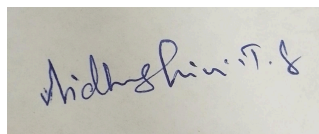
Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I Vidhushini.T.S (192511061) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6

Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5
Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

Answers :

1. Backtracking Search to find a valid assignment :

Given :

Doctors: D1, D2, D3

Shift:

- MorningShift
- Afternoon Shift
- Night Shift

Constraints:

- D1 cannot work Night.
- D2 must work before D3 (Afternoon).
- Only one doctor can be assigned to each shift.
- D3 cannot work Morning.
- Every doctor must be assigned exactly one shift.

Problem Solving:

★ Step 1: Define Variables and Domains

Doctor	Possible Shifts
D1	Morning, Afternoon
D2	Morning, Afternoon, Night
D3	Afternoon, Night

★ Step 2: Start Backtracking

Initially, Assignment = { }

★ Step 3: Assign D1

Try Morning

Assignment: {D1 = Morning}

Constraint Check:

- D1 is not Night.
- No duplicate shift.

Proceed.

★ Step 4: Assign D2

Morning is already assigned.

Try Afternoon

Assignment: {D1 = Morning, D2 = Afternoon}

Constraint Check:

- One doctor per shift.
- D2 can still be before D3.

Proceed.

★ Step 5: Assign D3

Morning is not allowed.

Afternoon is already occupied.

Try Night

Assignment: {D1 = Morning, D2 = Afternoon, D3 = Night}

Constraint Check:

- D3 is not Morning
- D2 works before D3 (Afternoon < Night)
- Only one doctor in each shift
- Every doctor assigned once

Solution found.

Constraint Verification :

Constraint	Status
D1 cannot work Night	Satisfied
D2 before D3	Satisfied
One doctor per shift	Satisfied
D3 cannot work Morning	Satisfied
Every doctor assigned exactly one shift	Satisfied

Final Valid Schedule

Shift	Assigned Doctor
Morning	D1
Afternoon	D2
Night	D3

2. Using Manhattan Distance Heuristic :

Grid:

S			X	
	X		X	
	X			
		X	X	
X				G

S = Start , G = Goal , X = Obstacle

Manhattan Distance Formula: $h(n) = |x_g - x| + |y_g - y|$

Initial State = (0,0) , Goal State = (4,4)

Step-by-Step Expansion:

1. Step 1:

Expand (0,0)

Possible Nodes:

- (1,0) $\rightarrow h = 7$
- (0,1) $\rightarrow h = 7$

Priority Queue : [(1,0), (0,1)]

2. Step 2:

Expand (1,0)

Add

(2,0) $\rightarrow h = 6$

Priority Queue : [(2,0), (0,1)]

3. Step 3:

Expand (2,0)

Add

(3,0) $\rightarrow h = 5$

Priority Queue : [(3,0), (0,1)]

4. Step 4:

Expand (3,0)

Add

(3,1) $\rightarrow h = 4$

Priority Queue : [(3,1), (0,1)]

5. Step 5:

Expand (3,1)

Add

$(4,1) \rightarrow h = 3$

Priority Queue : $[(4,1), (0,1)]$

6. Step 6:

Expand $(4,1)$

Add

$(4,2) \rightarrow h = 2$

Priority Queue : $[(4,2), (0,1)]$

7. Step 7:

Expand $(4,2)$

Add

$(4,3) \rightarrow h = 1$

Priority Queue : $[(4,3), (0,1)]$

8. Step 8:

Expand $(4,3)$

Add

$(4,4) \rightarrow h = 0$

Priority Queue : $[(4,4), (0,1)]$

9. Step 9:

Expand $(4,4)$

Goal reached.

Optimal Path :

$(0,0)$

|

$(1,0)$

|

$(2,0)$

|

$(3,0) \rightarrow (3,1)$

|

$(4,1) \rightarrow (4,2) \rightarrow (4,3) \rightarrow (4,4)$

Path Cost:

Each move = 1 , Total Moves = 8 , Optimal Cost = 8

3. Autonomous Rescue Robot in a Disaster-Affected Building :

Constraints:

- The robot can move only Up, Down, Left, and Right.
- Some cells are blocked (obstacles) and cannot be crossed.
- The robot can observe only the adjacent cells.
- Each movement has a cost of 1.
- Entering a risky zone adds an extra cost of 2.
- The robot must minimize the total cost while reaching the survivor.
- The robot updates its knowledge dynamically while moving.

(a) Analyze the Given Constraints and Explain Their Effect on Decision-Making:

The rescue robot must make intelligent decisions while navigating through the building. Each constraint directly affects how the robot selects its path.

1. Limited Movement (Up, Down, Left, Right)

The robot can move only in four directions. Diagonal movement is not allowed. Therefore, at every position the robot considers only four possible actions before making a decision.

Effect:

- Reduces the number of possible moves.
- Simplifies path planning.
- Ensures movement follows the building layout.

2. Obstacles

Some rooms are blocked because of debris or collapsed walls. These cells cannot be entered.

Effect:

- The robot must avoid blocked cells.
- If a path becomes blocked, another route must be searched.
- Sometimes the shortest path may not be available.

3. Partial Observability

The robot can observe only its neighboring cells. It does not know the complete layout of the building when it starts.

Effect:

- Decisions are made using only the available information.
- The robot continuously gathers new information while moving.
- The search plan may change whenever new obstacles are discovered.

4. Movement Cost

Every movement from one cell to another has a cost of 1.

Effect:

- The robot prefers paths requiring fewer movements.
- Unnecessary exploration is avoided.
- The overall travel cost is minimized.

5. Risk Zones

Certain areas may contain fire, unstable floors, smoke, or damaged structures. Entering these cells adds an extra cost of 2.

Effect:

- Risky areas are avoided whenever possible.
- The robot compares different paths before selecting one.
- A slightly longer but safer path may have a lower total cost than a shorter risky path.

6. Dynamic Knowledge Update

As the robot moves, it discovers new obstacles and safe paths.

Effect:

- The robot continuously updates its internal map.
- Previously planned paths may be changed.
- The robot adapts to changes in the environment.

Summary of Constraints

Constraint	Effect on Robot
Four-direction movement	Limits possible actions at each step
Obstacles	Forces the robot to find alternate routes
Partial observability	Decisions are made with incomplete information
Movement cost	Encourages shorter paths
Risk zones	Encourages safer paths with lower total cost
Dynamic updates	Allows replanning when new information is found

(b) Develop a Solution Using Uniform Cost Search (UCS) with an Online Search Agent:

Since the robot does not know the complete environment initially and must minimize total cost, Uniform Cost Search (UCS) combined with an Online Search Agent is an appropriate solution.

Working Steps:

Step 1: Start Position

The robot starts from the source position S.

Initially, only the surrounding cells are known.

Step 2: Sense the Environment

The robot scans all adjacent cells and identifies whether each cell is:

- Free
- Blocked
- Risky

The newly discovered information is stored in memory.

Step 3: Calculate Path Cost

For every possible move, the robot calculates the cumulative cost.

Example:

- Normal cell = 1
- Risky cell = $1 + 2 = 3$

The robot keeps track of the total cost from the start.

Step 4: Expand the Lowest-Cost Node

Among all possible paths, UCS always selects the node with the minimum accumulated cost.

This guarantees that the cheapest path discovered so far is explored first.

Step 5: Update the Map

As the robot moves, it may discover:

- New obstacles
- Additional risky zones
- Alternative safe paths

The internal map is immediately updated.

Step 6: Recalculate the Best Path

If the previously selected path becomes blocked or too costly, the robot recalculates the path using the updated information.

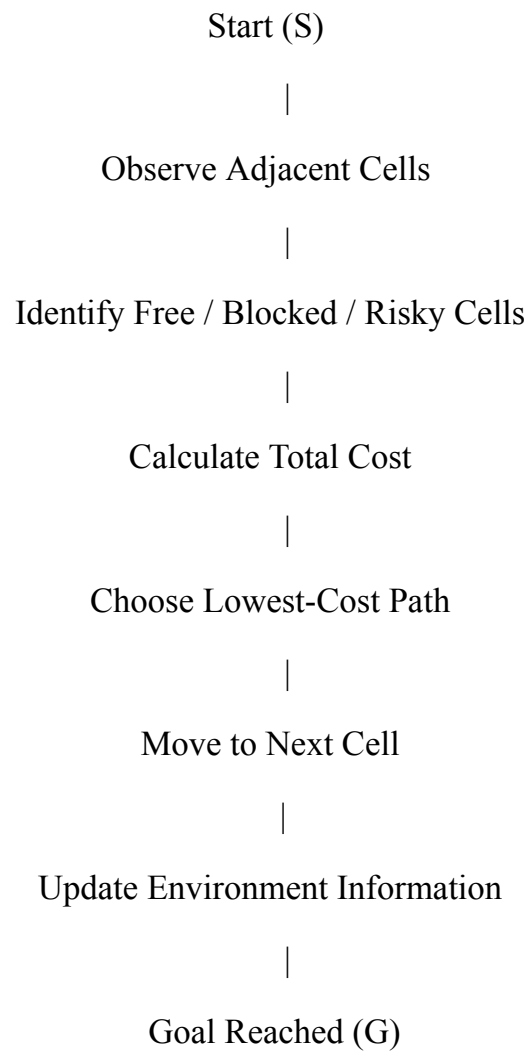
This makes the robot adaptive to changing environments.

Step 7: Reach the Goal

The process continues until the survivor (Goal G) is reached.

The selected path has the minimum total cost considering both movement and risk.

Flow Diagram:



(c) Demonstrate AI Concepts:

1. Intelligent Agent:

The rescue robot is an intelligent agent because it can:

- Sense the environment using sensors.
- Process available information.
- Make decisions independently.
- Perform actions to achieve the goal.

The robot follows the Sense → Think → Act cycle throughout the rescue operation.

2. Rationality:

A rational agent always chooses the action that maximizes its performance.

In this problem, the robot behaves rationally by:

- Avoiding obstacles.
- Avoiding risky zones whenever possible.
- Selecting the path with the minimum total cost.
- Replanning whenever better information becomes available.

Thus, every decision helps achieve the goal efficiently and safely.

3. Environment Type:

Environment Property	Description
Observability	Partially Observable because the robot can sense only adjacent cells.
Deterministic	The result of each movement is predictable.
Dynamic	The robot's knowledge changes as new areas are explored.
Sequential	Every decision affects future actions.
Discrete	The building is divided into separate grid cells.
Single Agent	Only one rescue robot performs the search.

(d) Justify the Chosen Approach:

Uniform Cost Search combined with an Online Search Agent is the most suitable approach for this problem for the following reasons:

- UCS always expands the path with the lowest cumulative cost, ensuring an optimal solution when costs vary.
- It considers both the normal movement cost and the additional cost of risky zones, leading to safer path selection.
- The Online Search Agent allows the robot to operate even without complete knowledge of the environment.
- The robot can update its map whenever new obstacles or risky areas are discovered and immediately modify its plan.
- This approach is efficient for real-world rescue operations where conditions are uncertain and may change over time.
- By combining cost optimization with continuous learning from the environment, the robot reaches the survivor safely while minimizing the overall travel cost.